

Comprehensive Biology Notes

Mitochondria - Detailed Study

Mitochondria are double-membrane organelles responsible for producing ATP through cellular respiration.

- 1 Outer membrane: Smooth and permeable
- 2 Inner membrane: Folded into cristae for increased surface area
- 3 Matrix: Contains enzymes, mitochondrial DNA, and ribosomes
- 4 Has its own DNA → supports Endosymbiotic Theory

Functions include ATP production, regulation of apoptosis, calcium storage, and heat generation.

Cell Organelles

- 1 Nucleus: Controls cell activities and stores DNA
- 2 Ribosomes: Protein synthesis
- 3 Endoplasmic Reticulum: Smooth (lipid synthesis), Rough (protein synthesis)
- 4 Golgi Apparatus: Modifies and packages proteins
- 5 Lysosomes: Digestion and waste removal

Photosynthesis (Detailed)

Occurs in chloroplasts in two stages:

- 1 Light Reaction: Produces ATP and NADPH
- 2 Dark Reaction (Calvin Cycle): Produces glucose

Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Cellular Respiration (Detailed)

- 1 Glycolysis: Occurs in cytoplasm, breaks glucose into pyruvate
- 2 Krebs Cycle: Occurs in mitochondria matrix
- 3 Electron Transport Chain: Produces maximum ATP

Genetics

- 1 DNA structure: Double helix
- 2 Genes: Functional units of heredity
- 3 RNA types: mRNA, tRNA, rRNA
- 4 Mutation: Changes in DNA sequence

Human Body Systems

- 1 Digestive System: Breaks down food
- 2 Respiratory System: Gas exchange
- 3 Circulatory System: Transports oxygen and nutrients
- 4 Nervous System: Controls body responses

Ecology

- 1 Ecosystem: Interaction of living and non-living components
- 2 Food Chain: Flow of energy
- 3 Food Web: Interconnected food chains
- 4 Biogeochemical Cycles: Carbon and nitrogen cycles

Evolution

Evolution is the gradual change in species over time.

- 1 Natural Selection: Survival of the fittest
- 2 Adaptation: Traits that improve survival
- 3 Speciation: Formation of new species